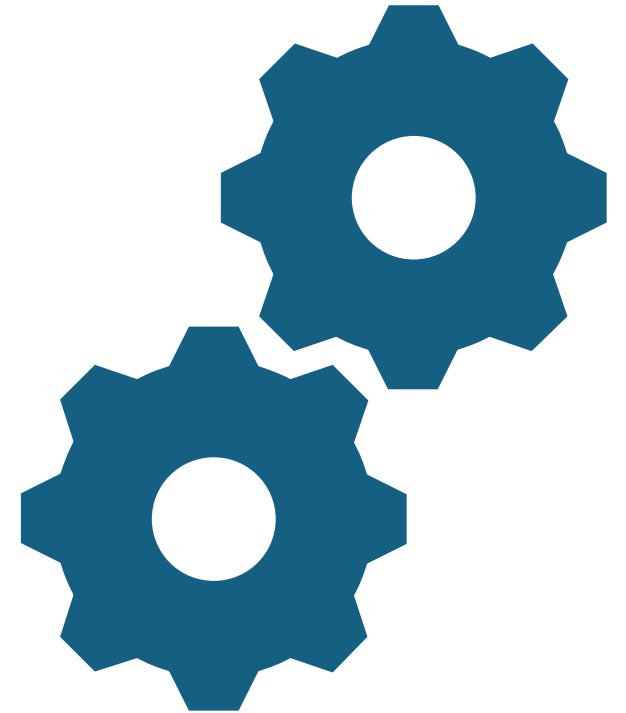


Advanced Deterministic Systems Optimization

ISE 555 Unit 2



Overview

- Part 1: Feasibility and Constraints
- Part 2: Optimality
- Part 3: Convex Functions and Convex Combinations
- Part 4: Optimization by moving in Feasible Directions

Part 1: Feasibility and Constraints



Feasibility and Constraints

- Feasibility is all about satisfaction of constraints. Any point that satisfies constraints is said to be *feasible*.
- Some math:
 - Let x represent the variables of the problem (assume there are N of them) We use x_i to represent individual variables
 - When variables are real, we say $x \in \mathbb{R}^N$ (but $x_i \in \mathbb{R}$)
 - When variables are integers, we say $x \in \mathbb{Z}^N$
 - When variables are binary, we say $x_i \in \{0,1\}$ or $x \in \mathbb{Z}^N$

Feasibility and Constraints (Cont.)

Functions $g_i(x)$ can represent constraints:

$$g_i(x) = 0, \quad i \in \mathcal{E} \quad (\mathcal{E} \text{ refers to the set of equality constraints})$$

$$g_i(x) \geq 0, \quad i \in \mathcal{I} \quad (\mathcal{I} \text{ refers to the set of inequality constraints})$$

Three points:

- 1) all equality constraints can be written in the form above
- 2) all equality constraints can be represented by 2 inequality constraints (or can reduce size N , how?)
- 3) all inequality constraints can be written in the form above

DEFINITION: A point x is a **feasible point** if x satisfies $g_i(x) = 0$ for all $i \in \mathcal{E}$ and x also satisfies $g_i(x) \geq 0$ for all $i \in \mathcal{I}$

DEFINITION: The set of all feasible points x is the **feasible region** or **feasible set**, which we call $\mathcal{S}$.

Part 2: Optimality

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What is Optimality?

- Standard form: $\min_{x \in \mathcal{S}} f(x)$ and we add constraints by saying “subject to” or “s.t.”
- Maximization vs. minimization: how does $\max_{x \in \mathcal{S}} f(x)$ become minimization?

DEFINITION: $x = x_*$ (some write x^*) is a **global minimizer** of $f(x)$ if

$$f(x_*) \leq f(x) \text{ for all } x \in \mathcal{S}$$

We call it a **strict global minimizer** if $<$ holds (instead of the more general $\leq$)

DEFINITION: $x = x_*$ (some write x^*) is a **local minimizer** of $f(x)$ if

$$f(x_*) \leq f(x) \text{ for all } x \in \mathcal{S} \text{ such that } \|x - x_*\| < \varepsilon$$

We call it a **strict local minimizer** if $<$ holds (instead of the more general $\leq$)

- When might you need global optimality and when might you need local?
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Part 3: Convex Functions and Convex Combinations

Convex Functions and Convex Combinations

- Convex combination: y is a **convex combination** of points x_i if

$$y = \sum_i x_i \alpha_i \text{ where } \sum_i \alpha_i = 1 \quad \text{and } \alpha_i \geq 0 \text{ for all } i$$

- Convex sets: $\mathcal{S}$ is a **convex set** if for any x and y in $\mathcal{S}$ we have

$$\lambda x + (1 - \lambda)y \in \mathcal{S} \text{ for all } 0 \leq \lambda \leq 1$$

- Convex functions: $f(x)$ is a **convex function** on $\mathcal{S}$ if it satisfies

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) \text{ for all } 0 \leq \lambda \leq 1 \text{ and any } x, y \in \mathcal{S}$$

- NOTES: If we replace $\leq$ with $<$ at all x then it is called **strictly convex**. If we replace $\leq$ with $\geq$ then it is **concave**.

WHITE BOARD FOR EXAMPLES

- The optimization of a convex objective function over a convex feasible set is called a **convex optimization problem**.

THEOREM: Any local optimizer of a convex optimization problem is also a global optimizer.

What about Calculus and Convexity?

- In calculus, we learned that a function is convex if and only if its second derivate is non-negative:

$$f(x) \text{ is convex if and only if } \frac{d^2 f(x)}{dx^2} \geq 0 \text{ for all } x$$

- This definition is consistent with the formal set definition. To see that, think of the set definition as meaning that the line segment connecting any two points on the function cannot have any other points in the function above it.
- This is effectively the notion of a convex function being a “cup” and thus any local minimum must be a global minimum.
- Note, there is a multi-dimensional version of the calculus definition:

If $x \in \mathbb{R}^N$, then $f(x)$ is convex if and only if $x^T M x \geq 0$ for all x where M is the matrix whose (i, j) -th entry is given by $M_{ij} = \frac{\partial^2 f(x)}{\partial x_i \partial x_j}$ (also known as the Hessian of the function)

Convexity of Objective Functions

- The optimization of a convex function over a convex feasible set is written mathematically as:

$$\begin{array}{ll}\text{minimize} & f(x) \\ \text{subject to} & g_i(x) \leq 0, \quad i = 1, \dots, m\end{array}$$

where $f(x)$ and $g_i(x)$ are all convex functions.

- Thus, finding problems with convex objectives is important if we want to claim global optimality!
- Convex or Concave? [HINT: Using derivatives sometimes helps]

- | | | |
|----|---|---------|
| 1. | $f(x) = -2x^2$ | CONCAVE |
| 2. | $f(x) = +2x^2$ | CONVEX |
| 3. | $f(x) = e^{ax}$ with $a \in \mathbb{R}$ | CONVEX |
| 4. | $f(x) = x^a$ with $0 \leq x \leq 1$ | CONCAVE |
| 5. | $f(x) = \max x_1, x_2, \dots, x_n$ | CONVEX |
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Part 1 (Cont.) : Feasibility and Constraints

Feasibility and Constraints (more definitions)

DEFINITION: At a specific feasible point $\bar{x}$, an inequality constraint $g_i(x) \geq 0$ is called **active** (or binding) if $g_i(\bar{x}) = 0$ and **inactive** (or nonbinding) if $g_i(\bar{x}) > 0$.

QUESTION: Can a feasible point be both active and inactive?

QUESTION: Can a feasible point be neither active nor inactive?

NOTE: All equality constraints must be active at any feasible point. Why?

DEFINITION: The **active set** at a feasible point is the set of all constraints that are active at that point.

DEFINITION: The **boundary** of the feasible set $\mathcal{S}$ is the set of all feasible points that contain at least one active inequality constraint

NOTE: All equality constraints are always active, so we do not consider them when defining the boundary

DEFINITION: The **interior points** are the points in the feasible set for which all of the inequality constraints are inactive

NOTE: We do not consider any active constraints when defining interior points (other than to be feasible)

Example from Lecture 1 (Woodshop/Cabinet mfg))

Resulting linear program was:

$$\begin{array}{ll}\text{maximize} & z = 100x_1 + 150x_2 + 200x_3 + 400x_4 \\ \text{subject to} & 10x_1 + 12x_2 + 25x_3 + 20x_4 \leq 5000 \\ & 2x_1 + 4x_2 + 8x_3 + 12x_4 \leq 1500 \\ & x_1, x_2, x_3, x_4 \geq 0\end{array}$$

QUESTIONS:

What constitutes a point x ?

What are the constraint functions $g_i(x)$?

What is the set $\mathcal{E}$ and the set $\mathcal{J}$?

What is the set $\mathcal{S}$ of feasible points?

Is point $x = (0, 125, 50, 50)^T$ feasible or infeasible?

If feasible which constraints are active and inactive? Is it an interior point?

Is point $x = (40, 100, 100, 50)^T$ feasible or infeasible?

If feasible which constraints are active and inactive? Is it an interior point?

Another Example

- Here is a set of constraints:
$$g_1(\mathbf{x}) = x_1 + 2x_2 + 3x_3 - 6 = 0$$
$$g_2(\mathbf{x}) = x_1 \geq 0$$
$$g_3(\mathbf{x}) = x_2 \geq 0$$
$$g_4(\mathbf{x}) = x_3 \geq 0$$

QUESTIONS: What is the set $\mathcal{E}$ and the set $\mathcal{I}$?

Are the following points feasible? If so, what constraints are active?

Is the point on the boundary or the interior?

(i) $\mathbf{x} = (0, 0, 2)^T$

(ii) $\mathbf{x} = (3, 0, 1)^T$

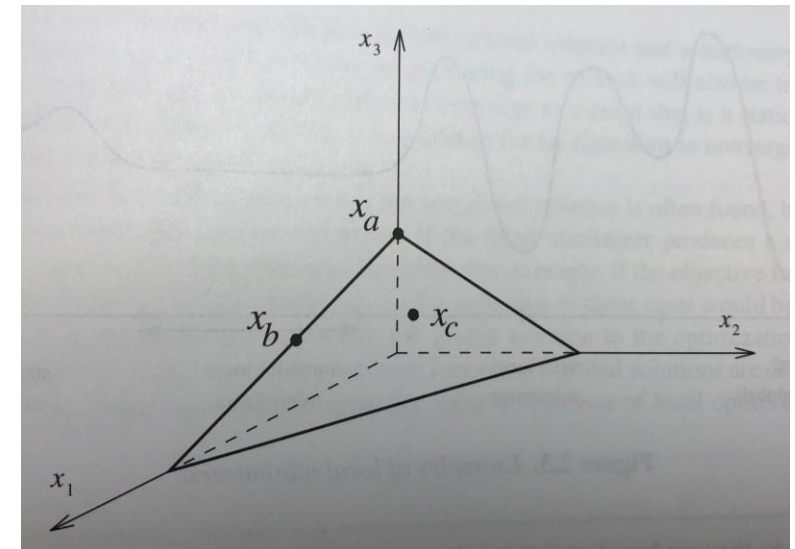
(iii) $\mathbf{x} = (0, 2, 1)^T$

(iv) $\mathbf{x} = (1, 1, 1)^T$

What is the set $\mathcal{S}$ of feasible points?

note $\mathbf{x}_a$ is (i), $\mathbf{x}_b$ is (ii) and $\mathbf{x}_c$ is (iv)

What is the boundary?



Part 4: Optimization by moving in Feasible Directions



“General” Optimization Algorithms

General Algorithm 1

- 1) Specify an initial guess x_0
- 2) For $k = 0, 1, \dots$ do the following:
 - i. If x_k is optimal, stop
 - ii. Determine x_{k+1} as a new (better) guess

General Algorithm 2

- 1) Specify an initial guess x_0
- 2) For $k = 0, 1, \dots$ do the following:
 - i. If x_k is optimal, stop
 - ii. Determine a search direction p_k
 - iii. Determine a step length α_k such that
$$x_{k+1} = x_k + \alpha_k p_k$$
 is a new (better) guess

Determining direction p_k

- p_k is referred to as the **descent direction** (for minimization)
- To minimize properly we need the next step to be an improvement for small step sizes α_k , thus:

$$f(x_k + \alpha p_k) < f(x_k) \text{ for } 0 < \alpha \leq \varepsilon$$

- Nonlinear programming methods of optimization are all about making good repeated
- For linear objective functions $f(x) = c^T x$ for some vector c , and we have

$$f(x_k + \alpha p_k) = c^T x_k + \alpha c^T p_k \quad \text{and} \quad f(x_k) = c^T x_k$$

- So the above condition becomes:

$$c^T x_k + \alpha c^T p_k < c^T x_k \quad \Rightarrow \quad c^T p_k < 0$$

DEFINITION: the vector p is a **feasible descent direction** at x_k for a linear programming problem with objective $f(x)$ and feasible set $\mathcal{S}$ if (for some $\varepsilon > 0$)

$$x_k + \alpha p \in \mathcal{S} \quad \text{and} \quad f(x_k + \alpha p) < f(x_k) \quad \text{for all } 0 < \alpha \leq \varepsilon$$

Feasible-point Methods

- Example: Objective $f(x)$ with one equality constraint $x_1 + x_2 = 1$
this is of the form $c_i^T x = b_i$ with $c_i = (1,1)^T$ and $b_i = 1$
if we move in direction p , then $c_i^T (x + \alpha p) = b_i$ only if $c_i^T p = 0$
or $p_1 + p_2 = 0$
or p is in the **null space** of c_i

What happens if we change it to inequality constraint $x_1 + x_2 \geq 1$?

For inactive feasible points x any p will work (for small steps)

For active feasible points we need $c_i p \geq 0$ to stay feasible

Conclusion is that the null space is important for finding feasible points around the boundary

Feasible-point Example

- Consider the following linear program:

$$\text{minimize } 2x_1 - x_2$$

$$\text{subject to } x_1 + x_2 \leq 2; \quad x_1, x_2 \geq 0$$

$$\text{NOTE: } f(x) = c^T x \text{ such that } c = (2, -1)$$

- Start with guess $x^{(0)} = (1, 1)$
- Determine feasible descent. Note $x^{(0)}$ is an active feasible point, so we need a direction $p^{(0)}$ such that $c^T p^{(0)} < 0$ while also meeting the active constraint.

That means $2p_1^{(0)} - p_2^{(0)} < 0$. We have $x_1 + x_2 \leq 2$ as the active constraint, so we also need $p_1^{(0)} + p_2^{(0)} \leq 0$.

- Try $p^{(0)} = (-1, 1)$. Now $x^{(1)} = x^{(0)} + \alpha p^{(0)} = (1 - \alpha, 1 + \alpha)$.
 - This means that:
 - $x_1^{(1)} + x_2^{(1)} = 2 \leq 2$ for any value of $\alpha > 0$
 - $x_1^{(1)} = 1 - \alpha \geq 0$ for $\alpha \leq 1$
 - $x_2^{(1)} = 1 + \alpha \geq 0$ for any value of $\alpha > 0$
 - So we can take $\alpha = 1$, and then $x^{(1)} = (0, 2)$ which is the optimal solution.
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Feasible-point Example – Try 2

- We needed a feasible descent that had $2p_1^{(0)} - p_2^{(0)} < 0$ and also had $p_1^{(0)} + p_2^{(0)} \leq 0$
 - What if we instead tried $p^{(0)} = (-1, 0)$. Now $x^{(1)} = x^{(0)} + \alpha p^{(0)} = (1 - \alpha, 1)$.
 - This means that:
 - $x_1^{(1)} + x_2^{(1)} = 2 - \alpha \leq 2$ for any value of $\alpha > 0$
 - $x_1^{(1)} = 1 - \alpha \geq 0$ for $\alpha \leq 1$
 - $x_2^{(1)} = 1 \geq 0$ for any value of $\alpha > 0$
 - So we can take $\alpha = 1$, and then $x^{(1)} = (0, 1)$ which is not the optimal solution. Although it is an improvement.
 - We now still have an active feasible point, but now the active constraint is $x_1 \geq 0$, so we need $p_1^{(1)} \geq 0$; and we still need $2p_1^{(1)} - p_2^{(1)} < 0$.
 - What if we tried $p^{(1)} = (0, 1)$ so we would get $x^{(2)} = x^{(1)} + \alpha p^{(1)} = (0, \alpha + 1)$.
 - Here we can move as much as $\alpha = 1$, this gets to $x^{(2)} = (0, 2)$ which is the optimal solution.
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Feasible Descent Summary

- There are two conditions for feasible descents:
 1. Must maintain feasibility (meet constraints)
 2. Must improve objective (descent)
 - The descent has two parts: a direction p and a magnitude α
 - The direction p needs be chosen to improve the objective and maintain any active constraints
 - The magnitude α needs to be chosen to meet all constraints – as large as possible is usually best (note that all active constraints should be already satisfied for any $\alpha > 0$)
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Summary

- Lots of math today – consider reviewing this PDF version on BrightSpace
 - These are the foundations of the algebra behind optimization, and we've focused on those that are most important to linear programming
 - Next week we will apply these to begin to derive the Simplex Method
 - HW #2 is due next week
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